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A PROJECT REPORT - 3/30/2026

Gamer

A PROJECT REPORT

on

"Fake Review Detection"

Submitted to

KIIT Deemed to be University

In Partial Fulfilment of the Requirement for the Award of

BACHELOR'S DEGREE IN

INFORMATION TECHNOLOGY

BY

Daksh Sinha — 2306021

Prabin Kumar Samal — 2306038

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UNDER THE GUIDANCE OF

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SCHOOL OF COMPUTER ENGINEERING

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

BHUBANESWAR, ODISHA - 751024

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CERTIFICATE

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Computer Science & Engineering OR Information Technology) at KIIT Deemed to be university, Bhubaneswar. This work is done during year 2025-2026, under our guidance.

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Project Guide

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ABSTRACT

The rapid growth of e-commerce and online review platforms has made customer reviews one of the most influential factors in consumer decision-making. However, the integrity of these reviews is increasingly compromised by fake or deceptive content, planted either to inflate a product's reputation or to damage a competitor's standing. This project addresses the challenge of automatically identifying such fraudulent reviews using machine learning techniques.

We propose a system that uses Natural Language Processing (NLP)-based text features to distinguish fake reviews from genuine ones. The dataset used is the Amazon Fine Food Reviews dataset, loaded as Reviews.csv, yielding 164,074 usable samples after dropping null entries. The labelling strategy assigns fake (label = 1) to reviews with Score ≤ 2 and genuine (label = 0) to reviews with Score > 2 , producing a perfectly balanced dataset of 82,037 samples per class. Two ensemble classifiers — Random Forest and XGBoost — are trained and compared on TF-IDF vectorized text features (max_features=5000, ngram_range=(1,2)).

Random Forest (n_estimators=200, max_depth=10) achieved an accuracy of 81.8% with a macro-averaged F1-score of 0.82. XGBoost (n_estimators=150, max_depth=4, learning_rate=0.1) achieved an accuracy of 82.1% with a macro-averaged F1-score of 0.82. Both models were evaluated on a stratified 20% test set comprising 32,815 samples (16,408 genuine, 16,407 fake). ROC curves and AUC scores confirm XGBoost as the marginally superior model across all classification thresholds. This work demonstrates that a TF-IDF-based text classification approach is effective for automated fake review detection and lays the groundwork for more sophisticated real-world deployments.

Keywords: Fake review detection, machine learning, TF-IDF, Random Forest, XGBoost, NLP, Amazon reviews, sentiment analysis, text classification, ensemble learning, opinion spam.

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FAKE REVIEW DETECTION

Chapter 1

Introduction

In the digital age, online reviews have become the modern word-of-mouth. Before making a purchase, booking a hotel, or choosing a restaurant, most people turn to platforms like Amazon, Yelp, or Google Reviews to gauge what others think. This dependency on peer feedback, while powerful, has introduced a significant vulnerability: fake reviews. Businesses, fraudsters, and reputation management agencies have discovered that artificial reviews

can sway consumer decisions and manipulate search rankings — making the detection of such content a critical challenge for both platforms and researchers.

Fake reviews can be broadly categorized into two types: positive fake reviews that artificially inflate a product's rating to boost sales, and negative fake reviews designed to harm a competitor's reputation. These reviews are often written without any genuine experience with the product and may be commissioned in bulk, making them difficult to detect manually at scale. According to industry estimates, a substantial fraction of reviews on major platforms may be fraudulent, raising urgent questions about consumer trust, market fairness, and platform integrity.

Traditional approaches to fake review detection relied heavily on manual content moderation or rule-based filtering. However, these methods are neither scalable nor adaptive enough to counter the evolving tactics of review spammers. Machine learning offers a more robust alternative by learning patterns from textual data that distinguish genuine reviews from deceptive ones.

This project proposes a machine learning-based system for fake review detection using the Amazon Fine Food Reviews dataset (164,074 usable samples after preprocessing). The system leverages TF-IDF vectorization of review text with bigram support. Two classifiers — Random Forest and XGBoost — are trained, evaluated, and compared. Random Forest achieved 81.8% accuracy and XGBoost achieved 82.1% accuracy, both evaluated on a stratified test set of 32,815 samples. The report is structured as follows: Chapter 2 reviews fundamental concepts and existing literature; Chapter 3 outlines the problem statement and system design; Chapter 4 details implementation and results; Chapter 5 covers standards adopted; and Chapter 6 concludes with limitations and future scope.

School of Computer Engineering, KIIT, BBSR1

FAKE REVIEW DETECTION

Chapter 2

Basic Concepts/ Literature Review

This chapter provides an overview of the key concepts, tools, and prior research relevant to the problem of fake review detection. Understanding these foundations is essential before examining the implementation details of the proposed system.

2.1 TF-IDF Vectorization

Term Frequency-Inverse Document Frequency (TF-IDF) is a statistical measure used to evaluate how important a word is to a document within a corpus. Unlike simple word counts, TF-IDF penalizes words that appear frequently across all documents (such as "the" or "and"), making it especially useful for identifying distinctive terms in text classification tasks. The TF component measures how frequently a term appears in a given document, while the IDF component down-weights terms that are common across the entire corpus. In this project, TF-IDF i

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 10 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.